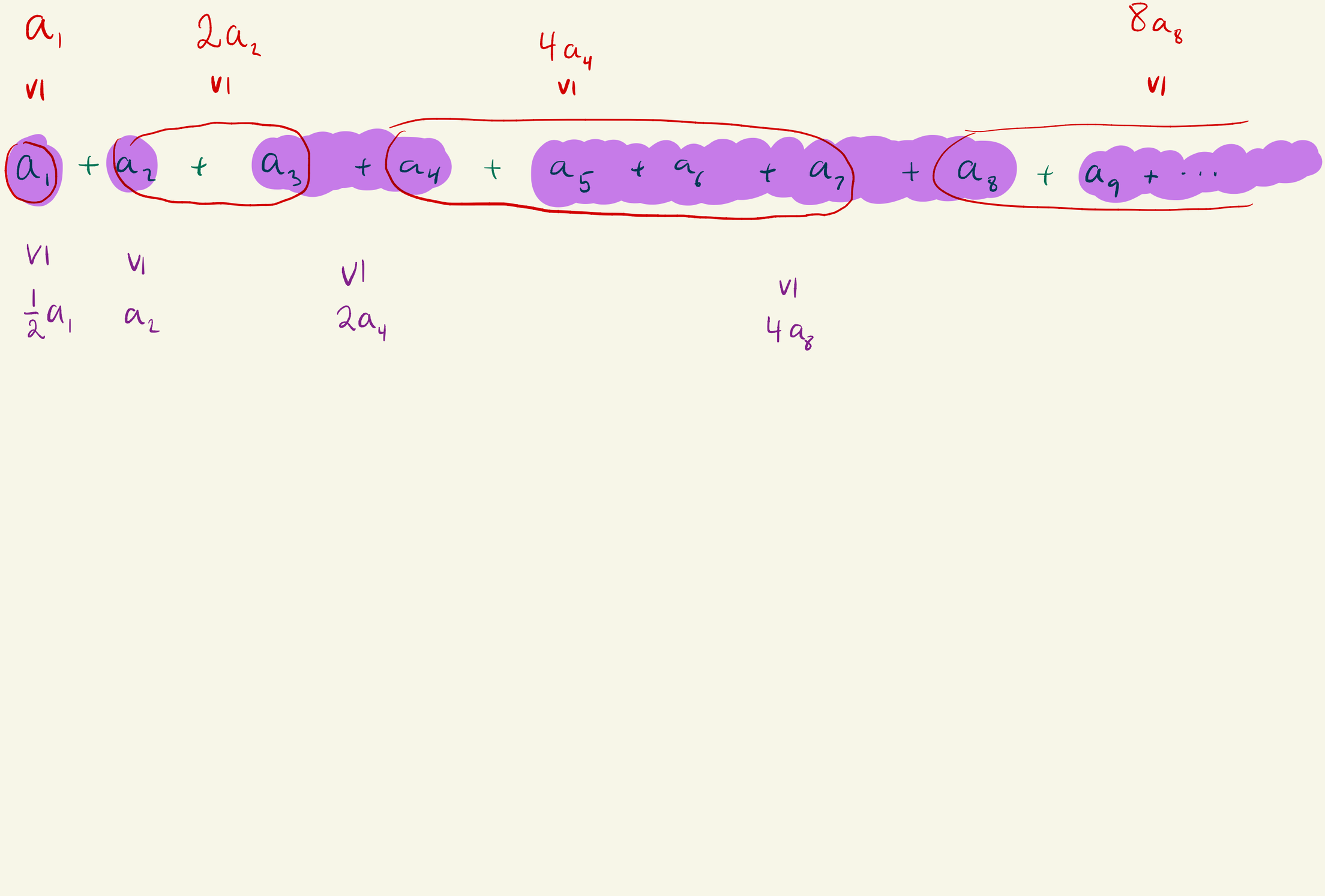
Topics

- Cauchy Condensation — Idea: Group terms
- Root test
- Ratio test

} Idea: Compair to Geometric Series.

Suppose a_n decreasing.



Thm (3.27) Suppose $a_1 \geq a_2 \geq \cdots \geq 0$. Then $\sum_{n=1}^{\infty} a_n$ converges iff

$$\sum_{k=0}^{\infty} 2^k a_{2^k} = a_1 + 2a_2 + 4a_4 + 8a_8 + \cdots \quad \text{converges.}$$

proof: Consider partial sums

$$s_n = a_1 + a_2 + \cdots + a_n$$

$$t_k = a_1 + 2a_2 + \cdots + 2^k a_k$$

For $n < 2^k$,

$$\begin{aligned} s_n &\leq a_1 + (a_2 + a_3) + \cdots + (a_{2^k} + \cdots + a_{2^{k+1}-1}) \\ &\leq a_1 + 2a_2 + \cdots + 2^k a_{2^k} \\ &= t_k \end{aligned}$$

For $n > 2^k$,

$$\begin{aligned} s_n &\geq a_1 + a_2 + (a_3 + a_4) + \cdots + (a_{2^{k-1}+1} + \cdots + a_{2^k}) \\ &\geq \tfrac{1}{2}a_1 + a_2 + 2a_4 + \cdots + 2^{k-1} a_{2^k} \\ &= \tfrac{1}{2} t_k \end{aligned}$$

(t_k) bounded $\Leftrightarrow$ (s_n) bounded so (t) conv. $\Leftrightarrow$ (s_k) conv.

Thm $\sum \frac{1}{n^p}$ converges if $p > 1$, and diverges if $p \leq 1$.

proof: If $p \leq 0$, $\frac{1}{n^p}$ does not go to 0, so series diverges.

Suppose $p > 0$.

Use thm 3.27:

$$\sum_{k=0}^{\infty} 2^k \frac{1}{(2^k)^p} = \sum_{k=0}^{\infty} 2^{k(1-p)} = \sum_{k=0}^{\infty} \left(2^{1-p}\right)^k$$

This converges iff $2^{1-p} < 1$ which happens iff $p > 1$.

In particular,

$\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Riemann Zeta function

$$\zeta(p) = \sum_{n=1}^{\infty} \frac{1}{n^p}$$

This formula makes sense for $p > 1$

Thm (3.17a) $\limsup s_n$ is a subsequential limit of (s_n).

Proof:

Let $E \subset \mathbb{R} \cup \{+\infty, -\infty\}$ be the set of subsequential limits of s_n. Recall, $\limsup_{n\to\infty} s_n = \sup E$.

- If $\sup E = \infty$, then (s_n) not bounded above, so there is a subsequence (s_{n_k}) s.t. $s_{n_k} \to \infty$.

- If $\sup E \in \mathbb{R}$ then $\sup E \in E$ because $E \cap \mathbb{R}$ is closed.

- If $\sup E = -\infty$, then for every $M \in \mathbb{R}$, there are fin. many $s_n > M$, otherwise there would be a subseq. lim in $[M, \infty]$, so $s_n \to -\infty$.

Thm (3.17b) $s^* = \limsup s_n$ is the unique subsequential limit s.t. $s > s^*$ implies $\exists N$ s.t. $\forall n \geq N$, $s_n < s$.

proof: Suppose $s > s^*$. Suppose there are ∞ many $s_n \geq s$

If (s_n) unbounded, ∞ is a subseq limit. If $s_n < b$ $\forall n$ then there is a real subseq limit $\geq s$ by cptness of $[s, b]$.

Thus, there are not ∞ many s_n greater than s.

Suppose not unique: s_1, s_2 are distinct subseq limits with this property. WLOG, $s_1 < s_2$. Let s be s.t. $s_1 < s < s_2$. There are ∞ many s_n greater than s, as s_2 is a subseq. limit, so s_1 does not have the property.

Thm (3.33) [Root test] Given a series $\sum a_n$,

Let $\alpha = \limsup \sqrt[n]{|a_n|}$

(a) $\alpha < 1 \Rightarrow \sum a_n$ converges

(b) $\alpha > 1 \Rightarrow \sum a_n$ diverges

(c) If $\alpha = 1$ $\sum a_n$ can converge or diverge.

Idea: $\sum a_n$ is roughly like the geometric series $\sum \alpha^n$

proof:

(a) Choose β s.t. $\alpha < \beta < 1$

Choose N s.t. $\sqrt[n]{|a_n|} < \beta \quad \forall n \geq N$

$n \geq N \Rightarrow |a_n| < \beta^n$

$\sum \beta^n$ conv. $\Rightarrow \sum a_n$ conv. by comparison test.

(b) Suppose $\alpha > 1$.

There exists subseq. (n_k) s.t. $\lim_{k\to\infty} \sqrt[n_k]{|a_{n_k}|} = \alpha$.

Thus $|a_n| > 1$ for infinitely many n, so (a_n) does not go to 0, so $\sum a_n$ diverges.

(c) What if $\alpha = 1$?

- $\sum_{n=0}^{\infty} \frac{1}{n}$ diverges — $\sqrt[n]{\frac{1}{n}} = \frac{1}{\sqrt[n]{n}} \to 1$
- $\sum_{n=0}^{\infty} \frac{1}{n^2}$ converges — $\frac{1}{\sqrt[n]{n^2}} = \frac{1}{(\sqrt[n]{n})^2} \to 1$

Example

$$\sum \frac{n^5}{3^n}$$

$$\sqrt[n]{\frac{n^5}{3^n}} = \frac{\sqrt[n]{n^5}}{3} \to \frac{1}{3} \Rightarrow \text{Convergence}$$

Thm (3.34) [Ratio Test] The series $\sum a_n$

(a) converges if $\limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$

(b) diverges if $\left|\frac{a_{n+1}}{a_n}\right| \geq 1$ for all $n > N_0$ for some $N_0 \in \mathbb{N}$.

Idea: Again, compare with geometric series.

Proof:

(a) If $\limsup \left|\frac{a_{n+1}}{a_n}\right| < 1$, then $\exists \beta < 1$ and N s.t.,

$n \geq N \Rightarrow \left|\frac{a_{n+1}}{a_n}\right| < \beta$.

For $n \geq N$

$$|a_n| \leq \beta |a_{n-1}| \leq \beta^2 |a_{n-2}| \leq \cdots \leq \beta^{n-N} |\beta_N|$$

$\sum \beta^{-N} |\beta_N| \cdot \beta^n$ converges $\Rightarrow$ $\sum a_n$ converges.

(b) If $|a_{n+1}| \geq |a_n|$ for all $n \geq N_0$, $a_n \to 0$ does not hold.

Suppose $\left|\frac{a_{n+1}}{a_n}\right| \to 1$. No conclusion.

- $\sum \frac{1}{n}$ diverges $\qquad \sum \frac{1}{n^2}$ converges

$\frac{n+1}{n} \to 1$ $\qquad \frac{n^2+1}{n^2} = 1 + \frac{1}{n^2} \to 1$

Example

$\sum \frac{4^n}{n!}$

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{4^{n+1}}{(n+1)!}\,\frac{n!}{4^n} = \frac{4}{n+1} \to 0 \quad \Rightarrow$ Convergence

Example

① $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$

$a_n = \frac{1}{2^n}$

$\sqrt[n]{a_n} = \frac{1}{2} \Rightarrow$ Convergence

$\frac{|a_{n+1}|}{|a_n|} = \frac{1}{2} \Rightarrow$ convergence

② $\frac{1}{2} + 1 + \frac{1}{8} + \frac{1}{4} + \frac{1}{32} + \frac{1}{16} + \cdots$

$\frac{b_{n+1}}{b_n} = 2, \frac{1}{8}, 2, \frac{1}{8}, 2, \ldots$

$\liminf \left|\frac{b_{n+1}}{b_n}\right| = \frac{1}{8}$, $\limsup \left|\frac{b_{n+1}}{b_n}\right| = 2$

Ratio test inconclusive.

Root Test gives convergence

$$b_{2n} = \frac{1}{2^{2n+1}}$$

$$b_{2n+1} = \frac{1}{2^{2n}}$$

$$\sqrt[2n+1]{|b_{2n}|} = \frac{1}{2^{2n/2n+1}} = \frac{1}{2^{1-\frac{1}{2n+1}}} = \frac{1}{2}\sqrt[2n+1]{2} \to \frac{1}{2}$$

$$\sqrt[2n]{|b_{2n+1}|} = \frac{1}{2^{2n-1/2n}} = \frac{1}{2^{1-\frac{1}{2n}}} = \frac{1}{2}\sqrt[2n]{2} \to \frac{1}{2}$$